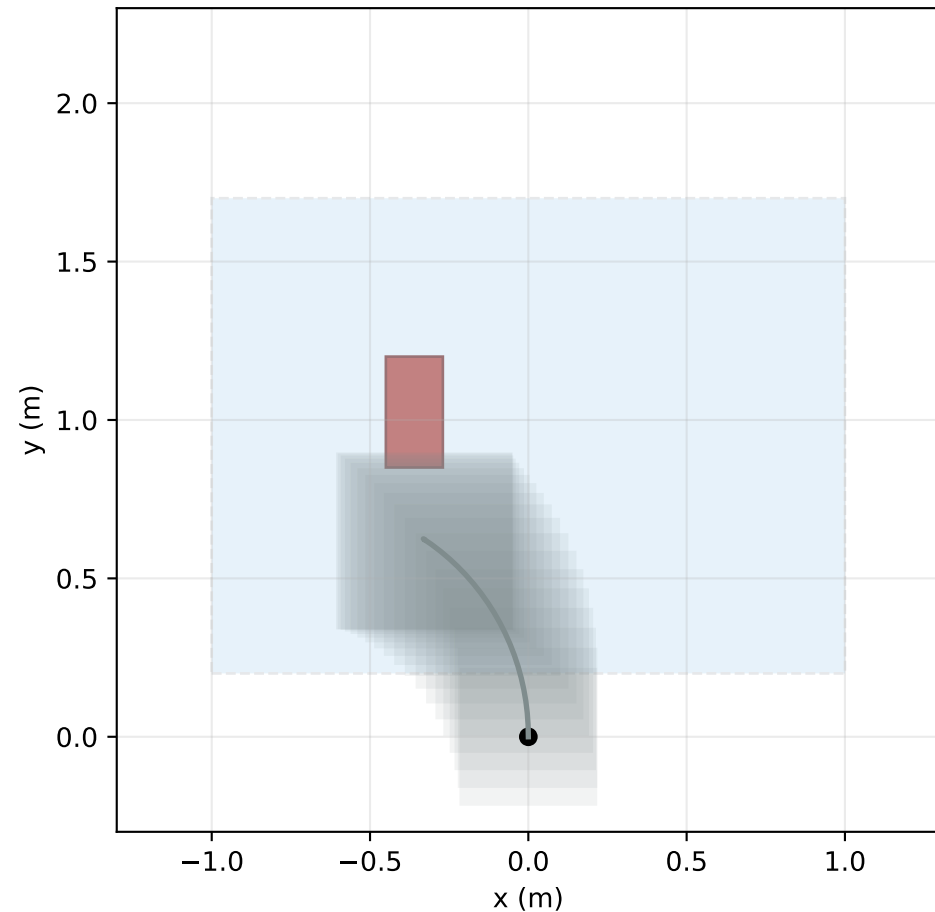
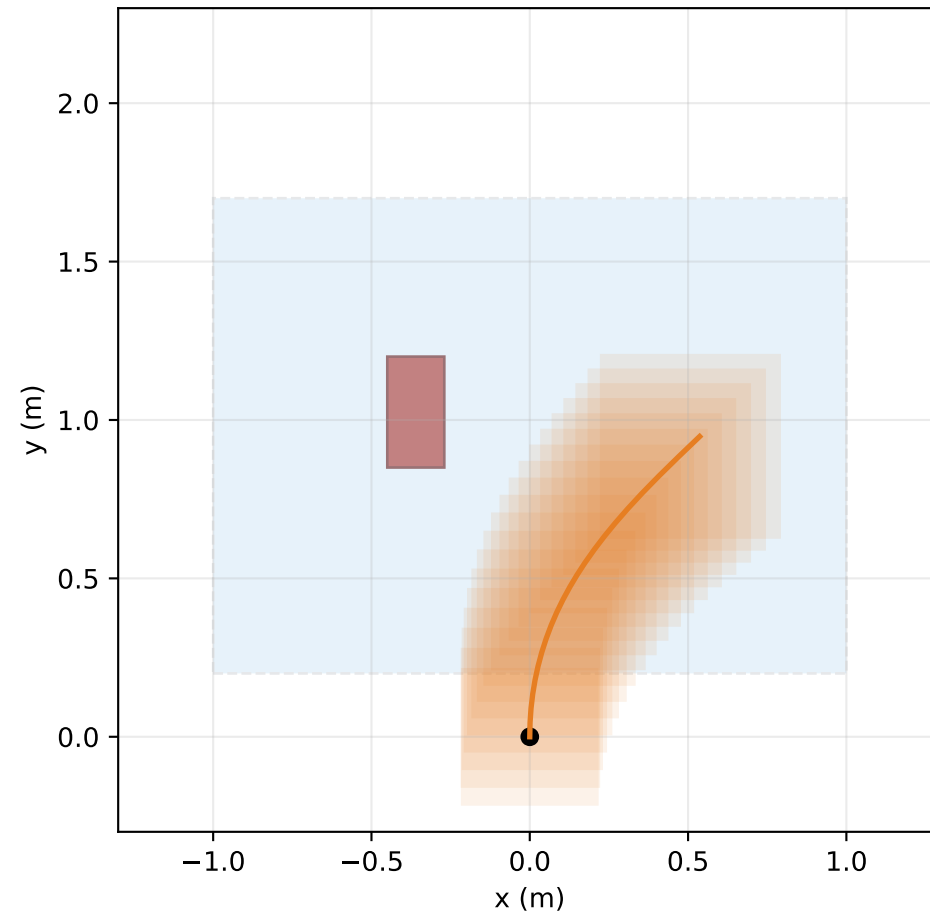


Planner-time disturbance awareness, scenario = tracking_error
each panel = world-space tube at the planner-picked k

goal-only (no MMR)
 $k = (1.0, 0.28)$, hard $m_{dir} = -0.046$, $m_{sym} = -0.046 \rightarrow$ UNSAFE



goal + symmetric MMR constraint
 $k = (-0.81, -1.0)$, hard $m_{dir} = +0.150$, $m_{sym} = +0.150 \rightarrow$ safe



goal + directional MMR constraint
 $k = (-0.81, -1.0)$, hard $m_{dir} = +0.150$, $m_{sym} = +0.150 \rightarrow$ safe

